Sowat Hossain Rafi

Cybersecurity and AI

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Professional Summary

Cybersecurity graduate with a background in computer science and engineering and internship experience developing Python features for AI video analysis. Built tools for cloud asset discovery and network intrusion detection, and investigated simulated security alerts using Splunk, TheHive and Wireshark. Co-authored an IEEE conference paper on deep learning for medical imaging.

Professional Experience

AI Engineer Intern
Unwind AI and Data Pty Ltd | Melbourne, Australia | Jun 2026–Sep 2026

- Contributed Python features to an AI video-analysis platform for physical security monitoring.
- Built and tested workflows for event detection and person re-identification.
- Worked with engineers and stakeholders to clarify requirements, review outputs and refine features.

Education

Master of Cybersecurity
La Trobe University | Melbourne, Australia | Mar 2024–Nov 2025

Bachelor of Science in Computer Science and Engineering
BRAC University | Dhaka, Bangladesh | Jan 2018–Jun 2022

Certifications and Professional Development

Security Analyst Level 1 (SAL1)
TryHackMe | Completed Nov 2026 | Valid through Nov 2029 | Verify on Credly

AI Security Level 1 (AI1)
TryHackMe | Completed Aug 2026 | Valid through Aug 2029 | Verify on Credly

Google Cybersecurity Professional Certificate V2
Google via Coursera | Verify on Credly

Cyber Incident Response and Digital Forensics
LinkedIn Learning | Instructor: Jason Dion

Technical Skills

Programming and automation: Python, Scapy, scikit-learn, Git, Bash and PowerShell.
Security operations: Splunk (SPL), TheHive, log analysis, alert triage and incident documentation.
Networking and forensics: Wireshark, Nmap, TCP/IP, DNS, HTTP/S and Windows Event Logs.
Cloud and asset discovery: AWS, Azure, Google Cloud and Active Directory inventory (projects).
Framework familiarity: MITRE ATT&CK, NIST Cybersecurity Framework and ISO/IEC 27001.

Technical Projects

Discovr Asset Discovery and Risk Triage
Independent project | Python, PySide6 and cloud SDKs | View on GitHub

- Developed a Python inventory tool that discovers assets across local networks, Active Directory, AWS, Azure and Google Cloud.
- Added asset classification, agent-capability indicators and risk ratings based on heuristics. Exported CSV, HTML and JSON reports to support security reviews.

Real-Time Network Intrusion Detection System
Independent prototype | Python, Scapy and scikit-learn | View on GitHub

- Built a Python prototype for live packet capture, TCP/IP flow analysis and alert logging.
- Implemented checks for SYN floods and port scans, used Isolation Forest for anomaly detection and generated structured JSON alerts.

Academic and Virtual Security Training

SOC Investigation and Digital Forensics
La Trobe University | Master's coursework | Mar 2025–Nov 2025

- Analysed logs in Splunk using SPL and triaged 15+ simulated phishing, brute-force and malware alerts in TheHive, checking indicators before escalation.
- Examined Windows Event Logs, disk images and Wireshark captures to reconstruct incident timelines and document evidence.
- Prepared reports on security posture, open risks and remediation for technical and executive audiences in a managed services simulation.

Malware Incident Response
Telstra via Forage | Virtual job simulation | Mar 2025

- Investigated a simulated malware incident and wrote a post-incident report on affected systems, root cause and recommended improvements to logging and detection.

Research Publication

Lossless Segmentation of Brain Tumors from MRI Images using 3D U-Net
Co-author, 2022 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE)
DOI 10.1109/CSDE56538.2022.10089263